

Missing Term in Series

Alphabetical series are combination of alphabets that are connected to each other with a rule. It might be increment, decrement, opposite and other mathematical operation. There are two types of Alphabetical series.

- (a) Alphabetical words and numerical based series.
- (b) String Based alphabetical series

Alphabetical word based series use some alphabetical words which are connected to each other. We have to find out their next term or any middle term. For example :

- (a) GIJ, LNO, ____, VXY, ACD

In above series first letter has been increased with + 5 place value, middle and last letter also increase with 5 place value.

String based series are combination of any type of strings means there is a alphabetical part that have any special rule and there is a fill in the blank within it. We have to find out that blank and fill according to following rules.

It may be of any type strings repeated, adding and dropping of letter also.

Some examples are given below.

- (a) bc abd / bc abd / bc abd / bc

It is a repeating part of bcabd string.

- (b) abc bcd cde def

It is adding and dropping based string, where we delete first letter and add new letter after 3rd letter.

Numerical series are based on Number and they will also follow any syntax., it may be increment, decrement, square, cube, prime number and any other combination of mathematical operation. Some examples are given below.

- (a) 71, 59, 48, 38, 29_____.

In above series there is a difference of - 12, - 11, - 10, - 9, - 8, So next term will be 21.

- (b) 5, 9, 18, 34, 59_____.

In above series they are having increment series as 2^2 , 3^2 , 4^2 , 5^2 , 6^2 , ... So next term is 95.



Points to Remember

- (A) In alphabetical series the letters are definitely related to their place value and opposites. A to Z sequence works as a normal sequence but it also performs in reverse sequence.
- (B) In numerical series the number having a particular sequence also uses some repeating rules which are :
 - (a) Increment and decrement in sequence of natural number
 - (b) Table of any number.
 - (c) Increment/decrement in prime / odd / even numbers.
 - (d) If there is a lot of difference between given numbers then they have to be related with square, cubes and multiplication of numbers.
 - (e) If upcoming results are having decimal then, it is an example of decimal numbers, mathematical operation and divisions.
- (C) In string based series always remember that string means repeating form of a combination. So we have to fill in the blanks according to given repeated form of combination.

Practice Set

1. 170, 149, 130, 113, ?
 (a) 97 (b) 98
 (c) 105 (d) 115
2. 96782345, 6782345, 678234, 78234, ?
 (a) 67814 (b) 7825
 (c) 7823 (d) 7884
3. PNPPONPP PNNPPNOOPN
 How many times letter 'P' is present in given responses which preceding by a consonant and followed by a vowel.
 (a) 5 (b) 2
 (c) 1 (d) none
4. 2 1 9 5 4 6 8 4 1 6 9 8 4 1 3
 How many odd number are present in the given series which is preceded by a perfect square and having followed by a odd number.
 (a) 2 (b) 3
 (c) 4 (d) 1
5. 158, 113, 78, 53, 28, ?
 (a) 13 (b) 12
 (c) 11 (d) 14
6. (Y,V,S,P,M,?)
 (a) J (b) L
 (c) K (d) R
7. 25, 216, 49, ?, 81, 1000
 (a) 625 (b) 512
 (c) 36 (d) 48
8. NOM, OPN, PQO, ?
 (a) RQP (b) OQP
 (c) QRP (d) RPO
9. M5D, O7G, Q10J, ?, U19P
 (a) S14M (b) S15N
 (c) N15S (d) N16S
10. 8, 17, 33, 67, ?
 (a) 131 (b) 132
 (c) 133 (d) 135
11. 100, 50, 52, 26, 28, ? 16, 8
 (a) 30 (b) 36
 (c) 14 (d) 32
12. Find the wrong term in the following series.
 60, 60, 61, 65, 74, ?
 (a) 74 (b) 90
 (c) 65 (d) 61
13. TOY, QPA, NQD, ?
 (a) RKH (b) HKM
 (c) KRH (d) KQN
14. BY, HS, NM, TG, ?
 (a) AZ (b) ZA
 (c) CX (d) WD
15. J, K, ?, P, T
 (a) N (b) M
 (c) O (d) P
16. Z, S, W, O, T, K, Q, G, ?, ?
 (a) N, C (b) N, D
 (c) O, C (d) O, D
17. 6, 11, 11, 20, 18, 31, 27, ?
 (a) 41 (b) 42
 (c) 43 (d) 44
18. b_abc_bca_c_bc in missing place term will be
 (a) caba (b) cabb
 (c) caaa (d) acad
19. 363, 297, ?, 198, 165, 143
 (a) 241 (b) 242
 (c) 243 (d) 96
20. l_n_mn_lm_n_lm
 (a) mlmlm (b) mlnml
 (c) mmlnl (d) lmlnl
21. 3, 15, 35, 63, 99, ?
 (a) 141 (b) 143
 (c) 151 (d) 169
22. 100, 80, 55, 25, -10, ?
 (a) -5 (b) -10
 (c) -15 (d) -50
23. GH, JKL, NOPQ, ?
 (a) STUVX (b) STUVW
 (c) STVUX (d) SUVTX
24. 62, 57, 68, 52, 74, 47, 80, ?, ?
 (a) 41, 83 (b) 42, 86
 (c) 43, 85 (d) 46, 88
25. ACE, BDF, GIK, -
 (a) HJL (b) MJH
 (c) JHM (d) HJI
26. WBP, SGM, OLJ, ?
 (a) LPG (b) MQG
 (c) NPH (d) KQG
27. XB, VD, TF, ?, PJ.
 (a) PQ (b) RH
 (c) RQ (d) RI
28. 151, 252, 454, ?, 1666
 (a) 585 (b) 888
 (c) 858 (d) 808

29. 300, 620, 1140, 1860, ?
 (a) 2580 (b) 2280
 (c) 2788 (d) 2780
30. 0, 2, 6, 12, ?, 30, 42
 (a) 24 (b) 20
 (c) 21 (d) 22
31. 5, 7, 11, 19, 35, 67, ?, 259
 (a) 130 (b) 129
 (c) 131 (d) 140
32. 3, 7, 15, 31, 63, 127, ?
 (a) 255 (b) 260
 (c) 245 (d) 265
33. Find out the number which is wrong according to given series.
 12, 77, 252, 620, 1292
 (a) 77 (b) 252
 (c) 620 (d) 1292
34. 7, 12, 22, 37, ?, 82, 112, 147
 (a) 50 (b) 58
 (c) 57 (d) 156
35. 7, 18, 36, ?, 93, 132
 (a) 92 (b) 83
 (c) 55 (d) 61
36. 826, 735, 644, 553, ?
 (a) 461 (b) 462
 (c) 564 (d) 584
37. 5, 11, 35, 143, -
 (a) 719 (b) 117
 (c) 917 (d) 722
38. a-bbc-aab-aa-abba-
 (a) cabaa (b) bacba
 (c) bbaaa (d) aabba
39. cc-dcn-cddc-n-ddcnn-d
 (a) cdndc (b) dnncc
 (c) dcndd (d) nccdn
40. Which of the following numbers is wrong in the series?
 3, 6, 10, 16, 21, 28
 (a) 16 (b) 10
 (c) 15 (d) 28
41. 10, 15, 35, 115, 435, ?
 (a) 1715 (b) 1517
 (c) 1716 (d) 1717
42. 12, 3, 24, 19, 44, ?
 (a) 43 (b) 46
 (c) 50 (d) 51
43. 17, ?, 65, 113, 129, 161, 209
 (a) 34 (b) 33
 (c) 40 (d) 44
44. 256, 25, 196, 49, ?, 81
 (a) 121 (b) 149
 (c) 144 (d) 16
45. 100, 70, 46, 28, ?, 10
 (a) 17 (b) 15
 (c) 20 (d) 16
46. 25, 96, 231, 448, ?
 (a) 765 (b) 760
 (c) 860 (d) 615
47. $\frac{5}{7}, \frac{8}{9}, ?, \frac{17}{21}, \frac{23}{37}$
 (a) $\frac{13}{12}$ (b) $\frac{12}{13}$
 (c) $\frac{15}{18}$ (d) $\frac{20}{17}$
48. 1495, 1494, 1485, 1460, ?, 1290, 1121
 (a) 1411 (b) 1114
 (c) 1322 (d) 1506
49. 6, 8, 11, 16, ?, 37, 58, 92
 (a) 20 (b) 21
 (c) 24 (d) 25
50. 5764801, 2401, 49, ?
 (a) 5 (b) 7
 (c) 8 (d) 9
51. 21, 42, 36, 72, 66, ?
 (a) 132 (b) 130
 (c) 169 (d) 225
52. 0.7, 1.2, 2.7, 5.2, ?, 13.2
 (a) 7.8 (b) 8.5
 (c) 8.7 (d) 14.5
53. 15, 18, 54, ?, 171, 174, 322
 (a) 159 (b) 62
 (c) 60 (d) 57
54. 100006, 10000, ?, 100, 16
 (a) 1005 (b) 1006
 (c) 1000 (d) 999
55. 6, 43, 304, 2133, ?
 (a) 14938 (b) 94134
 (c) 14532 (d) 23581
56. 4917, 4882, 4836, ?
 (a) 4564 (b) 6544
 (c) 4768 (d) 6845
57. C, D, F, I, M, ?
 (a) S (b) T
 (c) K (d) R
58. DMRC, ELSB, ?, GJUZ, HIVY
 (a) FKTA (b) KMTA
 (c) FLTA (d) FKTZ

59. IJK, ABC, LMN, DEF, ?
 (a) OP (b) GH
 (c) PQ (d) RS
60. M, P, O, -, -, T, S
 (a) S, T (b) R, Q
 (c) S, K (d) R, P
61. BHARTI, HIRBAT, ITBHRA, ?
 (a) TAHIBR (b) TAHIBS
 (c) THAIBR (d) THABIR
62. LI, LXXI, XCI, CXXI, ?
 (a) CLX (b) CLXI
 (c) CLI (d) ICL
63. Find out wrong term in series
 541, 514, 312, 743, 945, 816
 (a) 312 (b) 945
 (c) 816 (d) 541

Which set of letters when sequentially placed at the gaps in the given letter series shall complete it.

64. D_RCD_RCDM_CD_
 (a) MMRD (b) MMRC
 (c) MRMM (d) MMRM
65. B_OK_OW_OOK_OW_O
 (a) ONBNB (b) ONBMB
 (c) ONBBN (d) NOBBN
66. _AY_Z_YB_AY_Z
 (a) BZAZB (b) ZBAZB
 (c) ZBAZA (d) BZAZB
67. ba__b__am__aam_
 (a) ambaba (b) amabbb
 (c) ambabb (d) amabbb
68. ab_bb_aa_abb_a_babb
 (a) amama (b) ambam
 (c) ambbm (d) ambma
69. 26, 39, 58.5, ?, 131.625
 (a) 87.75 (b) 876.5
 (c) 987.5 (d) 872.8
70. 260, 345, 437, 536, ?, 755, 875
 (a) 445 (b) 642
 (c) 752 (d) 644
71. CMG, FPJ, ISM, ?
 (a) LVP (b) NVZ
 (c) NVY (d) LVZ
72. 22, 22, 55, 220, -, 8470
 (a) 1154 (b) 1210
 (c) 1145 (d) 1165
73. 291, 146, 73.5, 37.25, ?
 (a) 19.125 (b) 20.25
 (c) 19.15 (d) 119.10

74. 40250, 8050, 3220, ?, 257.6, 51.52
 (a) 464 (b) 666
 (c) 644 (d) 486
75. 3, 15, 72, 380, 2310, ?
 (a) 16215 (b) 16212
 (c) 16210 (d) 16201
76. 9, 10, ?, 69, 280, 1405
 (a) 18 (b) 20
 (c) 23 (d) 22
77. 1000000, 59049, 4096, ?, 36, 5
 (a) 169 (b) 361
 (c) 343 (d) 225
78. 34, 18, 10, ?
 (a) 8 (b) 5
 (c) 7 (d) 6
79. 4, 10, 22, 46, ?, 190
 (a) 56 (b) 16
 (c) 76 (d) 94
80. 9, 121, ?, 21841, 262081
 (a) 1681 (b) 981
 (c) 2516 (d) 1625

Distinct Questions

81. 2, 5, 26, ?
 (a) 675 (b) 674
 (c) 768 (d) 677
82. In the series 3, 9, 27, 81... What will be the 8th term.
 (a) 6891 (b) 6561
 (c) 5661 (d) 6231
83. 1, 2, 3, 6, 9, 18, -, 54
 (a) 18 (b) 27
 (c) 36 (d) 81
84. 7, 8, 11, 17, 27, ?
 (a) 42 (b) 43
 (c) 45 (d) 46
85. 2, 4, 6, 12, 22, ?
 (a) 48 (b) 40
 (c) 42 (d) 51
86. A, C, E, G, K, ?
 (a) L (b) K
 (c) M (d) P
87. 7, 18, ?, 236, 1184
 (a) 55 (b) 54
 (c) 51 (d) 58
88. BBE, BEF, BHI, ?, CFA
 (a) DBE (b) CBD
 (c) DCB (d) EBA
89. 4, 3.5, 5, ?, 19.5, 50.25
 (a) 9.5 (b) 8
 (c) 9 (d) 10.5

90. 198, 33, 6.6, 1.65, ?
 (a) 0.65 (b) 0.85
 (c) 8.9 (d) 0.55
91. 19, 22, 39, 103, 359, 1383
 (a) 22 (b) 103
 (c) 359 (d) 39
92. -1, 2, 7, ?, 23, 34, 47
 (a) 13 (b) 14
 (c) 12 (d) 15
93. 12, 36, ?, 200, 206, 1442, 1450
 (a) 40 (b) 41
 (c) 45 (d) 29
94. 84, 69, 83, ?, 82, 71
 (a) 73 (b) 74
 (c) 89 (d) 70
95. Find out wrong term in series
 2, 6, 18, 32, 50, 72, 98
 (a) 6 (b) 18
 (c) 32 (d) 50
96. JL, IG, NP, ?, RT
 (a) CE (b) EC
 (c) RD (d) FD
97. 55, 107, 317, ?, 6309, 37847
 (a) 1514 (b) 1428
 (c) 6312 (d) 1263
98. 8, 15, 42, ?, 580, 2915
 (a) 142 (b) 141
 (c) 158 (d) 402
99. 25 | 12 | 94, 9 | 1 | 95, 24 | 1 | 95, 8 | 2 | 95, ?
 (a) 22 | 2 | 95 (b) 25 | 2 | 95
 (c) 23 | 2 | 95 (d) 18 | 2 | 95
100. 5, 13, 31, 69, 147, ?
 (a) 305 (b) 308
 (c) 304 (d) 307

Previous year Questions (Memory Based)

1. $\frac{D}{5}, \frac{G}{9}, \frac{J}{14}, \frac{M}{20}, ?$
 (a) $\frac{Q}{26}$ (b) $\frac{O}{26}$
 (c) $\frac{Q}{27}$ (d) $\frac{P}{27}$
2. DWEV, FUGT, HSIR, ?
 (a) JKQP (b) JPQK
 (c) JQKP (d) JPKQ
3. EV, GT, JQ, ?
 (a) OP (b) LN
 (c) NM (d) MN
4. 313, 623, 933, 1243, ?
 (a) 1863 (b) 2173
 (c) 1553 (d) 2483
5. B2D, E3H, I4M, ?
 (a) N5R (b) N5T
 (c) N5S (d) N5Q
6. Which set of letters when sequentially placed at the gaps in the given letter series shall complete it?
 a c b - c e - f -
 (a) dde (b) cde
 (c) dee (d) ddg
7. QST __, QS __ R, Q __ TR, __ STR
 (a) SQTR (b) RTSQ
 (c) TRQS (d) TSRQ
8. AMV, FQX, KUZ, ?
 (a) PYB (b) OXA
 (c) NYB (d) MYB
9. 27, 32, 30, 35, 33, ?
 (a) 28 (b) 31
 (c) 36 (d) 38
10. 71, 59, 48, 38, 29, ?
 (a) 18 (b) 21
 (c) 20 (d) 12
11. a e b d _ f j g i _ k o l n _
 (a) c m h (b) c h m
 (c) c g m (d) c j l
12. 5255, 5306, ____, 5408, 5459
 (a) 5057 (b) 5357
 (c) 2257 (d) 5157
13. b 3 P, c 6 R, d 12 T, e 24 V, ?
 (a) f 48 X (b) f 46 X
 (c) f 48 W (d) g 48 X
14. $\frac{c}{6}, \frac{e}{10}, \frac{g}{14}, \frac{i}{18}, ?$
 (a) $\frac{k}{22}$ (b) $\frac{k}{11}$
 (c) $\frac{p}{22}$ (d) $\frac{p}{11}$
15. BDFH, IKMO, PRTV, ?
 (a) WYAC (b) WXYA
 (c) WXYZ (d) WYZA

16. 2, 65, 7, 59, 12, 53, ?, ?
 (a) 15, 42 (b) 17, 45
 (c) 17, 47 (d) 18, 48
17. 1, 2, 8, 33, 148, ?
 (a) 265 (b) 465
 (c) 565 (d) 765
18. 18, 22, 21, 20, 24, 18, ?
 (a) 27 (b) 25
 (c) 16 (d) 28
19. AJKTU, BILSV, CHMRW, DGNQX, ?
 (a) FEOYZ (b) EFOPY
 (c) EOFZA (d) EFOPZ
20. hgf, kji, n ? ?
 (a) lp (b) up
 (c) oq (d) ml
21. 210, 195, 175, 150, 120, ?
 (a) 75 (b) 85
 (c) 90 (d) 95
22. Which one set of letters when sequentially placed at the gaps in the given letter series shall complete it ?
 XY_KX_ZK_YZK XYZ ____
 (a) ZYXKX (b) ZYKXZ
 (c) ZKXYK (d) ZXYKZ
23. Which one set of letters when sequentially placed at the gaps in the given letter series shall complete it ?
 ac __ bd __ ce __df __ egh
 (a) d, f, g, e (b) b, d, c, a
 (c) d, g, f, e (d) d, e, f, g
24. $\frac{W}{S}, \frac{U}{O}, \frac{S}{K}, \frac{Q}{G}, ?$
 (a) P/R (b) C/O
 (c) R/J (d) O/C
25. HIII, IJJJ, JKKL, KLLM, LMMN, ?
 (a) LNNO (b) MNNP
 (c) NOOP (d) MNNO
26. 1, 1, 6, 6, 11, 11, 16, ? ?
 (a) 13, 11 (b) 16, 21
 (c) 17, 21 (d) 21, 16
27. 6341, 5432, _____, 3614
 (a) 4253 (b) 4614
 (c) 4532 (d) 4523
28. 4E, 8I, 13N, 19T, ____?
 (a) 26U (b) 26A
 (c) 262 (d) 25Y
29. h _ eg _ fegh _ eghfe _
 (a) gffh (b) hhgg
 (c) ffgh (d) fhfg
30. __ 011121 __ 11121 __ 111__
 (a) 1002 (b) 1102
 (c) 1012 (d) 1211
31. CFI, IKM, OPQ, ____?
 (a) UUU (b) UST
 (c) VUS (d) TUV
32. Series : $\frac{AB}{C}, \frac{ZY}{X}, \frac{DE}{F}, \frac{WV}{U}, \frac{GH}{I}, \text{---?---}$
 (a) $\frac{SR}{Q}$ (b) $\frac{TS}{R}$
 (c) $\frac{ST}{R}$ (d) $\frac{RS}{Q}$
33. 18, 25, 23, 30, ____?
 (a) 25 (b) 35
 (c) 28 (d) 38
34. 8, 29, 113, 449, ____?
 (a) 673 (b) 984
 (c) 1484 (d) 1793
35. Find the next two letters in the given series ?
 B C E H L ? ?
 (a) XY (b) MN
 (c) QW (d) OP
36. Which one set of letters when sequentially placed at the gaps in the given letter series shall complete it ?
 a__b__a__ __n__bb__abbn
 (a) abnabb (b) bnbban
 (c) bnbbna (d) babban
37. 3, 4, 7, 11, 18, 29, ____?
 (a) 31 (b) 39
 (c) 43 (d) 47
38. AGMSY, CIOUA, EKQWC, ____?____, IOUAG, KQWCI
 (a) GMSYE (b) FMSYE
 (c) GNSYD (d) FMYES
39. 975, 864, 753, 642, ____?
 (a) 431 (b) 314
 (c) 531 (d) 532
40. 8, 24, 12, ____?____, 18, 54
 (a) 28 (b) 36
 (c) 46 (d) 38
41. Which set of letters when sequentially placed at the gaps in the given letter series shall complete it ?
 ____a__aaaba__ __ba__ab__
 (a) abaaaa (b) abaaba
 (c) aababa (d) ababaa
42. a, r, c, s, e, t, g, ____?
 (a) x, z (b) u, i
 (c) w, y (d) v, b

43. (?), PSVYB, EHKNQ, TWZCF, ILORU

- (a) BEHKN (b) ADGJM
(c) SVYBE (d) ZCFIL

44. 0, 4, 18, 48, ?, 180

- (a) 58 (b) 144
(c) 84 (d) 100

45. 36, 28, 24, 22, ?

- (a) 18 (b) 19
(c) 21 (d) 22

46. 7, 9, 13, 21, 37, ?

- (a) 58 (b) 63
(c) 69 (d) 72

47. Which one set of letters when sequentially placed at the gaps in the given letter series shall complete it ?

ac_cab_baca__ aba__aca__

(a) acbcc

(b) aacbc

(c) babbb

(d) bcbba

Choose the correct alternative from the given ones that will complete the series :

48. ___? DREQ, GUHT, JXKW

- (a) EFRS (b) TGSF
(c) JWVI (d) AOBN

49. 56, 90, 132, 184, 248, ___?

- (a) 368 (b) 316
(c) 362 (d) 326

50. 1, 4, 10, 19, 31, ?

- (a) 46 (b) 50
(c) 55 (d) 43

Practice Set Solutions

1. (b); **Explanation :**

The pattern is -21, -19, -17, -15.....

So, missing term = 113 - 15 = 98.

2. (c); **Explanation :**

The digits are removed one by one from the beginning and the end in alternate order so as to obtain the subsequent terms of the series.

3. (c); Ans. (c) Only one P follows that condition.

4. (a); There are two combinations present are 195, 413. So answer is 2.

5. (a); the pattern is -45, -35, -25, ...

So, missing term = 28 - 15 = 13.

6. (a); The series consist of alternate letters in reverse order with three place value difference. Hence, next missing term is J.

7. (b); **Explanation :** The series consists of squares and cubes of consecutive natural number i.e., $5^2, 6^3, 7^2, 8^3, 9^2, 10^3, 11^2, \dots$ So, missing term = $8^3 = 512$.

8. (c); In every term, every letter (first, middle, last) increases with next letters. So missing term is QRP.

9. (a); **S 14 M**, Clearly, the first letters of the terms are alternate.

The last letter of each term is three steps ahead of the last letter of the preceding term.

Thus, the next term would be S 14 M. Hence, the answer is (a).

10. (c); **Explanation :** The pattern is $\times 2 + 1, \times 2 - 1, \times 2 + 1, \times 2 - 1, \dots$

so, missing term = $67 \times 2 - 1 = 133$

11. (c); The pattern is $\div 2, +2, \div 2, +2$, so next term is $28 / 2 = 14$.

12. (b); The pattern is $0, +(1)^2, +(2)^2, +(3)^2, +(4)^2$, so next term is 90.

13. (c); **Explanation :** In first letter decrement of three place values, increment of one place value in second letter and in third letters $+2, +3, +4$.

14. (b); **Explanation :** First letter increment with six place value and second letter decrease with six values.

15. (b); **Explanation :** Continuously increment of natural numbers in place value $\dots 1, 2, 3, 4, \dots$

16. (a); **Explanation :** The given sequence is a combination of two series : I, Z, W, T, Q, ? And II. S, O, K, G, ? In first series, letters are continuously decreasing by 3 and in second one there is decrement of 4.

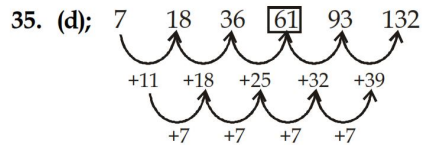
17. (d); It is a combination of 2 series, first series 6, 11, 18, 27, ... Which follow sequence as $+5, +7, +9, +11, \dots$ and second series is 11, 20, 31, ..., which follows sequence as $+9, +11, +13, \dots$ So next, missing term is $13 + 31 = 44$.

18. (a); The sequence is bcabca/bcabca/bcabca so missing term is caba.

19. (b); The pattern is -66, -55, -44, -33, -22. So missing term is $297 - 55 = 242$.

20. (b); The main sequence is lmn, increment of letter is reverse order. The sequence is lmn lmn lmn lmn So missing term is mlnml.

21. (b); The difference between digits are + 12, + 20, + 28, + 36, + 44, so missing term is $99 + 44 = 143$.
22. (d); The pattern is - 20, - 25, - 30, ..., So its missing term will be $- 10 - 40 = - 50$.
23. (b); Consecutive letter no. as +2, +3, +4, +5,..., (no of letter) and upcoming term is increase by place value of +2. So missing term is STUVW.
24. (b); Combination of 2 series and then 1st series is 62, 68, 74, 80, ..., having -6 difference and 2nd series is 57, 52, 47, ..., having -5 so missing term will be 42 and 86.
25. (a); There is oneplace value increment, in first option it follows.
26. (d); In given sequence first letter is decrease by 4, second letter is increased with five place value and 3rd letter decrease with 3 digit.
27. (b); The sequence is -2, +2, -2, +2. So missing terms is RH.
28. (c); The sequence is +101, +202, +404, +808... So, missing term is 858.
29. (d); The sequence is +320, +520, +720,... So, missing term is 2780.
30. (b); Here the respective difference between the terms is as follows :
2, 4, 6, 8,
Therefore, $12 + 8 = 20$
which is the required term.
31. (c); Here the respective difference between the term is as follows :
2, 4, 8, 16, 32, 64,
Therefore, the required term will be $67 + 64 = 131$
32. (a); Here on adding 1 to the double of the first term we get the next term.
As required- $3 \times 2 + 1 = 7$
 $7 \times 2 + 1 = 15$
 $15 \times 2 + 1 = 31$
 $31 \times 2 + 1 = 63$
 $63 \times 2 + 1 = 127$
 $127 \times 2 + 1 = 255$
33. (c); The term follow as $(x^4 - 4)$ where x run as 2, 3, 4,...
So, wrong term is 620.
34. (c); Here, the respective difference between terms is as follows : 5, 10, 15, 20, 25
As, $7 + 5 = 12$
 $12 + 10 = 22$
 $22 + 15 = 37$
 $\therefore$ the next term will be $= 37 + 20 = 57$



36. (b); The difference in 1st, 2nd and 3rd digit are follow -1, +1, -1 as respectively.
37. (a); The sequence is $\times 2 + 1, \times 3 + 2, \times 4 + 3, \times 5 + 4...$, So term is 719.
38. (a); Here the letter series is as follows-
acb bca aab baa aab baa
39. (c); Here the letter series is as follows-
ccd dcn ccd dcn ndd dcn ndd
40. (a);
-
- 3, 6, 10, 16, 21, 28
+3, +4, +5, +6, +7
+1, +1, +1, +1
41. (a); The sequence is $+5, + (5 \times 4), + (20 \times 4), + (80 \times 4), + (320 \times 4)$. So term is 1715
42. (a); Two series are present one is 12, 24, 44, ... having difference is 12, 20 so next term is 70. Second is 3, 19, 43, ... having difference + 16, + 24 so term is 43.
43. (b); The sequence is +16, +32, +48, +16, +32, ... So term is 33.
44. (c); The sequence is $16^2, 5^2, 14^2, 7^2, 12^2, 9^2$. So terms is 144.
45. (d); The sequence is, $-(6 \times 5), -(6 \times 4), -(6 \times 3), -(6 \times 2)$. So term is 16.
46. (a); The sequence is $[(5)^2 - 0], (10^2 - 2^2), (16^2 - 5^2), (23^2 - 9^2), (31^2 - 14^2)$
47. (b); In upper side sequence is +3, +4, +5, +6. ...on denominator the sequence is 2, 4, 8, 16.
48. (a); The sequence is $-1^2, -3^2, -5^2, -7^2, -11^2, -13^2$. So term is 1411.
49. (c); The sequence is +2, +3, +5, +8, +13. So term is 24.
50. (b); $\sqrt{5764801} = \sqrt{2401} \Rightarrow 49$ So, $\sqrt{49} = 7$.
51. (a); The sequence is $\times 2, -6, \times 2, -6, \times 2, -6$. So term is 132.
52. (c); The sequence is +0.5, +1.5, +2.5, +3.5. So term is 8.7
53. (d); The sequence is +3, $\times 3$, +3, $\times 3$. So, term is 57.
54. (b); The unit place replacement by 6 and alternate decrement of digits so term is 1006.
55. (a); The sequence is $(\times 7 + 1), (\times 7 + 3), (\times 7 + 5)$. So term is

56. (c); The sequence is $\overset{+11}{-35}, \overset{+22}{-46}, -68$. So term is 4768.
57. (d); The sequence is 3, 4, 6, 9, 13, 18 with difference of +1, +2, +3. So change them into alphabets then term is R.
58. (a); The sequence in alphabet is +1, -1, +1, -1 So term is FKTA.
59. (a); Here is combination of 3 set and their consecutive increment present here. So there is OP.
60. (b); The sequence is +3, -1, +3, -1, +3, -1. So term is R, Q.
61. (a); In this sequence word's letters are arranged in 264135 order. So ITBHRA changed in 264135 sequence then term will be TAHIBR.
62. (c); The given format is roman format of 51, 71, 91, 111, 131. So next term is CLI.
63. (c); Here the sum of unit digit and tenth digit is the 100th digit.
As, $1 + 4 = 5, 4 + 1 = 5$
 $2 + 1 = 3, 3 + 4 = 7$
 $5 + 4 = 9$, But in 816
 $6 + 1 = 7$ not 8.
64. (d); The sequence is DMRC | DMRC | DMRC | DMRC. So missing term is MMRM
65. (a); The sequence is Book Now / Book Now / Bo. So missing term ONBNB.
66. (b); ZA YB / ZAYB / ZAYB. So missing term is ZBAZB.
67. (c); The sequence is baamb / baamb / baamb. So sequence is ambab.
68. (d); The sequence is ababbma / ababbma / ababb. So missing term is ambma.
69. (a); The pattern is $\times 1.5, \times 1.5, \times 1.5, \times 1.5$. So missing term is 87.75.

70. (b); The pattern is $\overset{+7}{+85}, \overset{+7}{+92}, \overset{+7}{+99}, \overset{+7}{+106}, \overset{+7}{+113}, \overset{+7}{+120}$.
So missing term is 642.

71. (a);
- | | | | | | | |
|----|--------------------|----|--------------------|----|--------------------|----|
| 3 | $\xrightarrow{+3}$ | 6 | $\xrightarrow{+3}$ | 9 | $\xrightarrow{+3}$ | 12 |
| C | | F | | I | | L |
| | | | | | | |
| 13 | $\xrightarrow{+3}$ | 16 | $\xrightarrow{+3}$ | 19 | $\xrightarrow{+3}$ | 22 |
| M | | P | | S | | V |
| | | | | | | |
| 7 | $\xrightarrow{+3}$ | 10 | $\xrightarrow{+3}$ | 13 | $\xrightarrow{+3}$ | 16 |
| G | | J | | M | | P |

72. (b); The pattern is $\times 1, \times 2.5, \times 4, \times 5.5, \times 7$. So $55 \times 4 = 220$, then missing term is 1210.

73. (a); The pattern is $\left(\times \frac{1}{2} + 0.5\right), + \left(\times \frac{1}{2} + 0.5\right), + \left(\times \frac{1}{2} + 0.5\right)$. So missing term is 19.125.

74. (c); The patter is $+5, +2.5, +5, +2.5$. So missing term is 644.

75. (b); The sequence is $(\times 3, +2 \times 3), (\times 4, +3 \times 4), (\times 5, +4 \times 5)$. So, missing term is 16212.

76. (d); The pattern is add one in previous term and then multiplication with natural numbers. Example:
 $(a+1) \times 1 = 10, (10+1) \times 2 = 22, (22+1) \times 3 = 69$
So missing term is 22.

77. (c); The pattern is $10^6, 9^5, 8^4, 7^3, 6^2, 5^1$. So missing term is 343.

78. (d);
- | | | | |
|---------------------|--------------------|--------------------|---|
| 34 | 18 | 10 | 6 |
| $\xrightarrow{-16}$ | $\xrightarrow{-8}$ | $\xrightarrow{-4}$ | |
| | $\xrightarrow{+2}$ | $\xrightarrow{+2}$ | |

79. (d);
- | | | | | | |
|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------|-----|
| 4 | 10 | 22 | 46 | 94 | 190 |
| $\xrightarrow{\times 2+2}$ | $\xrightarrow{\times 2+2}$ | $\xrightarrow{\times 2+2}$ | $\xrightarrow{\times 2+2}$ | $\xrightarrow{\times 2+2}$ | |

80. (a); The pattern is $(\times 15 - 14), (\times 14 - 13), (\times 13 - 12), (\times 12 - 11)$. So missing term is 1681.

Distinct Solutions

81. (d); **Explanation** : Each term in the series is obtained by adding 1 to the square of the preceding term.
So, missing term = $(26)^2 + 1 = 677$.

82. (b); **Explanation** : Clearly, $3 \times 3 = 9, 9 \times 3 = 27, 27 \times 3 = 81, \dots$. So, the series is in G.P. in which $a = 3, r = 3$. Therefore 8th term = $ar^{8-1} = ar^7 = 3 \times 3^7 = (3 \times 2187) = 6561$.

83. (b); **Explanation** : The pattern is $\times 2, \times 3/2, \times 2, \times 3/2, \times 2, \dots$. So, missing term = $18 \times 3/2 = 27$.

84. (a); **Explanation** : The pattern is +1, +3, +6, ... i.e. +1, $+(1+2), +(1+2+3), \dots$. So, missing term = $27 + (1+2+3+4+5) = 42$.

85. (b); **Explanation** : The sun of any three consecutive terms of the series gives the next term, So, missing number = $6 + 12 + 22 = 40$.

86. (c); Alphabetical series are having place value as 1, 2, 3, 5, 7, 11, 13 are prime numbers. So missing terms is M.

87. (d); The pattern is $\times 2 + 4, \times 3 + 4, \times 4 + 4, \times 5 + 4 \dots$ So missing term is $18 \times 3 + 4 = 58$.

88. (b); The alphabet is changed with their place value then these are 225, 256, 289, 324,..., Which are square value of 15, 16, 17, 18,. So, next term is CBD.
89. (c); The sequence is $\times 0.5 + 1.5$, $\times 1 + 1.5$, $\times 1.5 + 1.5$. So, missing term is 9.
90. (d); The sequence is $\div 6$, $\div 5$, $\div 4$, $\div 3$. So missing term is 0.55
91. (a); The sequence is $+2^2$, $+2^4$, $+2^6$, $+2^8$, ... So, second term is 23 not 22.
92. (b); Here the series is as follows :
 $1^2 - 2 = -1$ $4^2 - 2 = 14$
 $2^2 - 2 = 2$ $5^2 - 2 = 23$
 $3^2 - 2 = 7$ $6^2 - 2 = 34$
93. (a); The sequence is $\times 3$, $+4$, $+5$, $+6$, $\times 7$, $+8$. So term is 40.
94. (d); The sequence is -15 , $+14$, -13 , $+12$, -11 . So term is 70.
95. (a); On looking from the end, we find that the difference between the numbers is respectively 26, 22, 18, 14, 10, 6. Therefore, 8 should come in place of 6.
96. (b); The sequence is -1 , $+5$, -9 , $+13$ in first letter and in second letter, sequence is -5 , $+9$, -13 , $+17$. So next missing term is EC.
97. (d); The pattern is $\times 2 - 3$, $\times 3 - 4$, $\times 4 - 5$, $\times 5 - 6$, $\times 6 - 7$. So missing term is 1263.
98. (b); The pattern is $(\times 1 + 7 \times 1)$, $(\times 2 + 6 \times 2)$, $(\times 3 + 5 \times 3)$. So missing term is 141.
99. (c); There is 15 days difference in between two given data.
100. (a); The sequence is $+8$, $+(8 + 10)$, $+(18 + 20)$, $+(38 + 40)$, $+(78 + 80)$ So term is 305.

Previous year Solutions (Memory Based)

1. (d); $D \xrightarrow{+3} G \xrightarrow{+3} J \xrightarrow{+3} M \xrightarrow{+3} P$
 $\bar{5} \xrightarrow{+4} \bar{9} \xrightarrow{+5} \bar{14} \xrightarrow{+6} \bar{20} \xrightarrow{+7} \bar{27}$
2. (c); $D \xrightarrow{+2} F \xrightarrow{+2} H \xrightarrow{+2} J$
 $W \xrightarrow{-2} U \xrightarrow{-2} S \xrightarrow{-2} Q$
 $E \xrightarrow{+2} G \xrightarrow{+2} I \xrightarrow{+2} K$
 $V \xrightarrow{-2} T \xrightarrow{-2} R \xrightarrow{-2} P$
3. (c); The gap between E,G,J is $+2$, $+3$. Similarly the gap between V, T, Q is -2 , -3 . So, new group will be $+4$ from J and -4 from Q. Hence, it will be NM.
4. (c); $313 \xrightarrow{+310} 623 \xrightarrow{+310} 933 \xrightarrow{+310} 1243 \xrightarrow{+310} 1553$
5. (c); $B \xrightarrow{+3} E \xrightarrow{+4} I \xrightarrow{+5} N$
 $2 \xrightarrow{+1} 3 \xrightarrow{+1} 4 \xrightarrow{+1} 5$
 $D \xrightarrow{+4} H \xrightarrow{+5} M \xrightarrow{+6} S$
6. (a); $a \xrightarrow{+1} c \xrightarrow{+1} b \xrightarrow{+1} d \xrightarrow{+1} c \xrightarrow{+1} e \xrightarrow{+1} d \xrightarrow{+1} f \xrightarrow{+1} e$
7. (b); $QST \xrightarrow{+1} R \xrightarrow{+1} QS \xrightarrow{+1} T \xrightarrow{+1} R \xrightarrow{+1} Q \xrightarrow{+1} S \xrightarrow{+1} TR \xrightarrow{+1} Q \xrightarrow{+1} STR$
8. (a); $A \xrightarrow{+5} F \xrightarrow{+5} K \xrightarrow{+5} P$
 $M \xrightarrow{+4} Q \xrightarrow{+4} U \xrightarrow{+4} Y$
 $V \xrightarrow{+2} X \xrightarrow{+2} Z \xrightarrow{+2} B$
9. (d); $27 \xrightarrow{+5} 32 \xrightarrow{-2} 30 \xrightarrow{+5} 35 \xrightarrow{-2} 33 \xrightarrow{+5} 38$
10. (b); $71 \xrightarrow{-12} 59 \xrightarrow{-11} 48 \xrightarrow{-10} 38 \xrightarrow{-9} 29 \xrightarrow{-8} 21$
11. (b); $aebd \xrightarrow{+1} c \xrightarrow{+1} f \xrightarrow{+1} i \xrightarrow{+1} h \xrightarrow{+1} k \xrightarrow{+1} l \xrightarrow{+1} m$
12. (b); $5255 \xrightarrow{+51} 5306 \xrightarrow{+51} 5357 \xrightarrow{+51} 5408 \xrightarrow{+51} 5459$
13. (a); $b \xrightarrow{+1} c \xrightarrow{+1} d \xrightarrow{+1} e \xrightarrow{+1} f$
 $3 \xrightarrow{\times 2} 6 \xrightarrow{\times 2} 12 \xrightarrow{\times 2} 24 \xrightarrow{\times 2} 48$
 $P \xrightarrow{+2} R \xrightarrow{+2} T \xrightarrow{+2} V \xrightarrow{+2} X$

14. (a); $c \xrightarrow{+2} e \xrightarrow{+2} g \xrightarrow{+2} i \xrightarrow{+2} k$
 $6 \xrightarrow{+4} 10 \xrightarrow{+4} 14 \xrightarrow{+4} 18 \xrightarrow{+4} 22$

15. (a); $B \xrightarrow{+7} I \xrightarrow{+7} P \xrightarrow{+7} W$
 $D \xrightarrow{+7} K \xrightarrow{+7} R \xrightarrow{+7} Y$
 $F \xrightarrow{+7} M \xrightarrow{+7} T \xrightarrow{+7} A$
 $H \xrightarrow{+7} O \xrightarrow{+7} V \xrightarrow{+7} C$

16. (c);
$$\begin{array}{ccccccc} & & -6 & & -6 & & -6 \\ & & \downarrow & & \downarrow & & \downarrow \\ 2 & 65 & 7 & 59 & 12 & 53 & \boxed{17} \quad \boxed{47} \\ \uparrow & & \uparrow & & \uparrow & & \uparrow \\ & +5 & & +5 & & +5 & \end{array}$$

17. (d); $1 \times 1 + (1)^2 = 1 + 1 = 2;$
 $2 \times 2 + (2)^2 = 4 + 4 = 8;$
 $8 \times 3 + (3)^2 = 24 + 9 = 33;$
 $33 \times 4 + (4)^2 = 132 + 16 = 148;$
 $148 \times 5 + (5)^2 = 740 + 25 = 765$

18. (a);
$$\begin{array}{ccccccc} & & -2 & & -2 & & \\ & & \downarrow & & \downarrow & & \\ 18 & 20 & 21 & 20 & 24 & 18 & \boxed{27} \\ \uparrow & & \uparrow & & \uparrow & & \uparrow \\ & +3 & & +3 & & +3 & \end{array}$$

19. (b); $A \xrightarrow{+1} B \xrightarrow{+1} C \xrightarrow{+1} D \xrightarrow{+1} E$
 $J \xrightarrow{-1} I \xrightarrow{-1} H \xrightarrow{-1} G \xrightarrow{-1} F$
 $K \xrightarrow{+1} L \xrightarrow{+1} M \xrightarrow{+1} N \xrightarrow{+1} O$
 $T \xrightarrow{-1} S \xrightarrow{-1} R \xrightarrow{-1} Q \xrightarrow{-1} P$
 $U \xrightarrow{+1} V \xrightarrow{+1} W \xrightarrow{+1} X \xrightarrow{+1} Y$

20. (d); $h \xrightarrow{+3} k \xrightarrow{+3} n$
 $g \xrightarrow{+3} j \xrightarrow{+3} m$
 $f \xrightarrow{+3} i \xrightarrow{+3} l$

21. (b);
$$\begin{array}{ccccccc} 210 & 195 & 175 & 150 & 120 & 85 \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ -15 & -20 & -25 & -30 & -35 & \end{array}$$

22. (a); $XY \boxed{Z} K/X \boxed{Y} ZK /$
 $\boxed{X} YZK / XYZ \boxed{K} / \boxed{X}$

23. (d); $a c \boxed{d} / b d \boxed{e} / c e \boxed{f} / d f \boxed{g} / e g h$

24. (d); $W \xrightarrow{-2} U \xrightarrow{-2} S \xrightarrow{-2} Q \xrightarrow{-2} O$
 $S \xrightarrow{-4} O \xrightarrow{-4} K \xrightarrow{-4} G \xrightarrow{-4} C$

25. (d); Obviously, the next term will be MNNO.

26. (b);
$$\begin{array}{ccccccc} 1, & 1 & 6, & 6 & 11, & 11 & 16, \boxed{16} \quad \boxed{21} \\ \uparrow & & \uparrow & & \uparrow & & \uparrow \\ & +5 & & +5 & & +5 & \end{array}$$

27. (d); $6 \xrightarrow{-1} 5 \xrightarrow{-1} 4 \xrightarrow{-1} 3$
 $3 \xrightarrow{+1} 4 \xrightarrow{+1} 5 \xrightarrow{+1} 6$
 $4 \xrightarrow{-1} 3 \xrightarrow{-1} 2 \xrightarrow{-1} 1$
 $1 \xrightarrow{+1} 2 \xrightarrow{+1} 3 \xrightarrow{+1} 4$

28. (b); $4 \xrightarrow{+4} 8 \xrightarrow{+5} 13 \xrightarrow{+6} 19 \xrightarrow{+7} \boxed{26}$
 $E \xrightarrow{+4} I \xrightarrow{+5} N \xrightarrow{+6} T \xrightarrow{+7} \boxed{A}$

29. (d); $h \boxed{f} eg / \boxed{h} fe g / h \boxed{f} eg / hfe \boxed{g}$

30. (a); $\boxed{1} 01112 / 1 \boxed{0} 1112 / 1 \boxed{0} 111 \boxed{2}$

31. (a); $C \xrightarrow{+6} I \xrightarrow{+6} O \xrightarrow{+6} U$
 $F \xrightarrow{+5} K \xrightarrow{+5} P \xrightarrow{+5} U$
 $I \xrightarrow{+4} M \xrightarrow{+4} Q \xrightarrow{+4} U$

32. (b);
$$\begin{array}{ccc} \overline{AB} & \xrightarrow{\quad} & \overline{DE} \xrightarrow{\quad} \overline{GH} \\ C & & F \quad I \\ \uparrow & & \uparrow \\ & +3 & +3 \end{array}$$

$$\begin{array}{ccc} \overline{ZY} & \xrightarrow{\quad} & \overline{WV} \xrightarrow{\quad} \overline{TS} \\ X & & U \quad R \\ \uparrow & & \uparrow \\ & -3 & -3 \end{array}$$

33. (c);
$$\begin{array}{ccccccc} & & +5 & & \\ & & \downarrow & & \\ 18 & 25 & 23 & 30 & \boxed{28} \\ \uparrow & & \uparrow & & \uparrow \\ & +5 & & +5 & \end{array}$$

34. (d);
$$\begin{array}{ccccccc} 8 & 29 & 113 & 449 & 1793 \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ +21 & +84 & +336 & +1344 \\ \uparrow & \uparrow & \uparrow & \uparrow \\ \times 4 & \times 4 & \times 4 & \times 4 \end{array}$$

35. (c); $B \xrightarrow{+1} C \xrightarrow{+2} E \xrightarrow{+3} H$
 $H \xrightarrow{+4} L \xrightarrow{+5} Q \xrightarrow{+6} W$

36. (b); $a \boxed{b} b \boxed{n} / a \boxed{bb} n / \boxed{a} bb \boxed{n} / abbn$

37. (d); $3 + 1 = 4; \quad 3 + 4 = 7;$
 $4 + 7 = 11; \quad 7 + 11 = 18$
 $11 + 18 = 29; \quad 18 + 29 = 47$

38. (a); $A \xrightarrow{+2} C \xrightarrow{+2} E \xrightarrow{+2} G \xrightarrow{+2} I \xrightarrow{+2} K$
 $G \xrightarrow{+2} I \xrightarrow{+2} K \xrightarrow{+2} M \xrightarrow{+2} O \xrightarrow{+2} Q$
 $M \xrightarrow{+2} O \xrightarrow{+2} Q \xrightarrow{+2} S \xrightarrow{+2} U \xrightarrow{+2} W$
 $S \xrightarrow{+2} U \xrightarrow{+2} W \xrightarrow{+2} Y \xrightarrow{+2} A \xrightarrow{+2} C$
 $Y \xrightarrow{+2} A \xrightarrow{+2} C \xrightarrow{+2} E \xrightarrow{+2} G \xrightarrow{+2} I$

39. (c); $975 \xrightarrow{-111} 864 \xrightarrow{-111} 753 \xrightarrow{-111} 642 \xrightarrow{-111} \boxed{531}$

40. (b); $8 \xrightarrow{\times 3} 24 \xrightarrow{+2} 12 \xrightarrow{\times 3} \boxed{36} \xrightarrow{+2} 18 \xrightarrow{\times 3} 54$

41. (a); $a a b a / a a b a / a a b a / a a b a$

42. (b); There are two alternative series :

$a \xrightarrow{+2} c \xrightarrow{+2} e \xrightarrow{+2} g \xrightarrow{+2} i$

$r \xrightarrow{+1} s \xrightarrow{+1} t \xrightarrow{+1} u$

Therefore, ? = ui

43. (b); $P \xrightarrow{+3} S \xrightarrow{+3} V \xrightarrow{+3} Y \xrightarrow{+3} B$
 $E \xrightarrow{+3} H \xrightarrow{+3} K \xrightarrow{+3} N \xrightarrow{+3} Q$
 $T \xrightarrow{+3} W \xrightarrow{+3} Z \xrightarrow{+3} C \xrightarrow{+3} F$
 $I \xrightarrow{+3} L \xrightarrow{+3} O \xrightarrow{+3} R \xrightarrow{+3} U$

Now, $P \xrightarrow{+4} T, E \xrightarrow{+4} I$

Therefore, the first letter of the first term should be

$E \xrightarrow{-4} A$

$A \xrightarrow{+3} D \xrightarrow{+3} G \xrightarrow{+3} J \xrightarrow{+3} M$

44. (d); $0 \xrightarrow{+4} 4 \xrightarrow{+10} 14 \xrightarrow{+16} 30 \xrightarrow{+22} 52 \xrightarrow{+28} 80 \xrightarrow{+34} 114$
 $4 \xrightarrow{+14} 18 \xrightarrow{+20} 38 \xrightarrow{+26} 64 \xrightarrow{+32} 96 \xrightarrow{+38} 134$
 $18 \xrightarrow{+20} 38 \xrightarrow{+26} 64 \xrightarrow{+32} 96 \xrightarrow{+38} 134$
 $30 \xrightarrow{+22} 52 \xrightarrow{+28} 80 \xrightarrow{+34} 114$
 $52 \xrightarrow{+28} 80 \xrightarrow{+34} 114$
 $80 \xrightarrow{+34} 114$

45. (c); $36 \xrightarrow{-8} 28 \xrightarrow{-4} 24 \xrightarrow{-2} 22 \xrightarrow{-1} \boxed{21}$

46. (c); $7 \xrightarrow{\times 2} 14 \xrightarrow{+4} 18 \xrightarrow{\times 2} 36 \xrightarrow{+8} 44 \xrightarrow{\times 2} 88 \xrightarrow{+16} 104 \xrightarrow{\times 2} 208 \xrightarrow{+32} 240$

47. (b); $ac \boxed{a} c / ab \boxed{a} b / a c a$
 $\boxed{c} / aba \boxed{b} / a c a \boxed{c}$

48. (d); $A \xrightarrow{+3} D \xrightarrow{+3} G \xrightarrow{+3} J$
 $O \xrightarrow{+3} R \xrightarrow{+3} U \xrightarrow{+3} X$
 $B \xrightarrow{+3} E \xrightarrow{+3} H \xrightarrow{+3} K$
 $N \xrightarrow{+3} Q \xrightarrow{+3} T \xrightarrow{+3} W$

49. (d); $56 \xrightarrow{+8} 64 \xrightarrow{+10} 74 \xrightarrow{+12} 86 \xrightarrow{+14} 100 \xrightarrow{+16} 116$
 $64 \xrightarrow{+10} 74 \xrightarrow{+12} 86 \xrightarrow{+14} 100 \xrightarrow{+16} 116$
 $74 \xrightarrow{+12} 86 \xrightarrow{+14} 100 \xrightarrow{+16} 116$
 $86 \xrightarrow{+14} 100 \xrightarrow{+16} 116$
 $100 \xrightarrow{+16} 116$

50. (a); The difference between the consecutive terms is 3, 6, 9, 12, 15.